Auteur: Thijs Van Schel
Studierichting: Informatica
Oefeningenreeks 4A nr. 15.9

$$\int \frac{x^3}{e^{2x}} dx$$

$$\left\{ \begin{array}{l} u = x^3 \\ dv = \dfrac{dx}{e^{2x}} \end{array} \right. \Rightarrow \left\{ \begin{array}{l} du = 3x^2 dx \\ v = -\dfrac{1}{2e^{2x}} \end{array} \right.$$

$$= \frac{-x^3}{2e^{2x}} + \frac{3}{2} \int \frac{x^2}{e^{2x}} dx$$

$$\left\{ \begin{array}{l} u = x^2 \\ dv = \dfrac{dx}{e^{2x}} \end{array} \right. \Rightarrow \left\{ \begin{array}{l} du = 2x dx \\ v = -\dfrac{1}{2e^{2x}} \end{array} \right.$$

$$= \frac{-x^3}{2e^{2x}} + \frac{3}{2} \left[\frac{-x^2}{2e^{2x}} + \int \frac{x}{e^{2x}} dx \right]$$

$$\left\{ \begin{array}{l} u = x \\ dv = \dfrac{dx}{e^{2x}} \end{array} \right. \Rightarrow \left\{ \begin{array}{l} du = dx \\ v = -\dfrac{1}{2e^{2x}} \end{array} \right.$$

$$= \frac{-x^3}{2e^{2x}} + \frac{3}{2} \left[\frac{-x^2}{2e^{2x}} + \left(\frac{-x}{2e^{2x}} + \frac{1}{2} \int \frac{1}{e^{2x}} dx \right) \right]$$

$$= \frac{-x^3}{2e^{2x}} - \frac{3x^2}{4e^{2x}} - \frac{3x}{4e^{2x}} - \frac{3}{8e^{2x}} + C$$

References